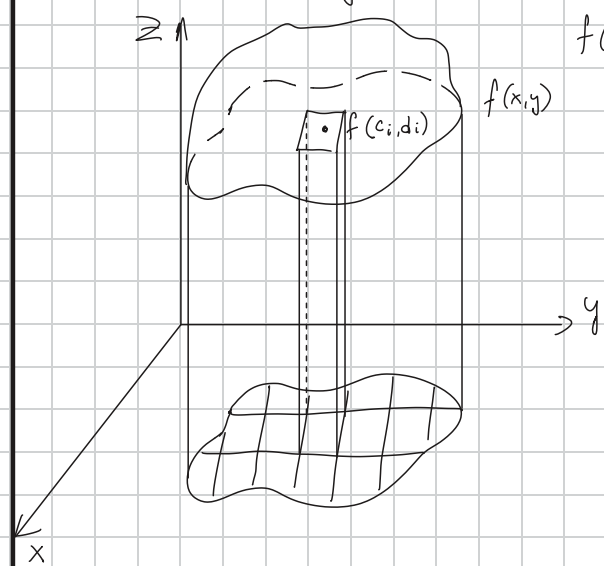


Вычисление двойного интеграла



$$V \approx \sum V_i = \sum \Delta x_i \cdot \Delta y_i \cdot f(c_i, d_i)$$

$$V = \lim \sum f(c_i, d_i) \Delta x_i \Delta y_i$$

Обозначим $\iint_D f(x, y) dx dy$

Формула

$$\iint_D f(x, y) dx dy = \int_a^b \left(\int_{y_1(x)}^{y_2(x)} f(x, y) dy \right) dx$$

глобальной *крайней*

Замечание

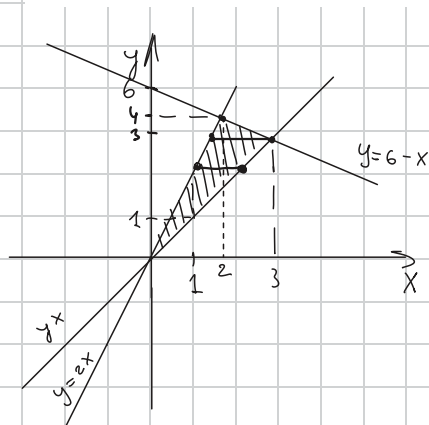
Правый (крайний) интеграл можно без скобок:

$$\int_a^b \left(\int_{y_1(x)}^{y_2(x)} f(x, y) dy \right) dx \xrightarrow{\text{замечание}} \int_a^b dx \int_{y_1(x)}^{y_2(x)} f(x, y) dy$$

вычисляем только

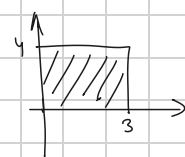
$$\iint_D f(x, y) dx dy \xrightarrow{y=c} \int_c^a dy \int_{x_1(y)}^{x_2(y)} f(x, y) dx$$

3493. Треугольник со сторонами $y = x$, $y = 2x$ и $x + y = 6$.



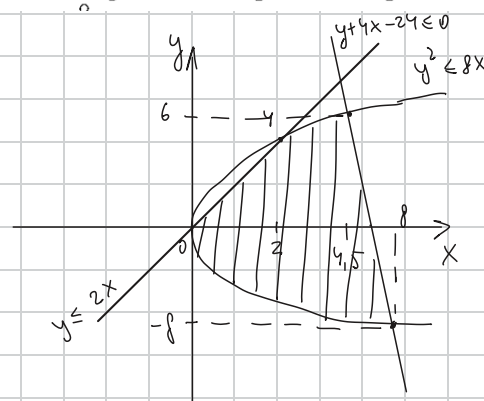
$$\int_0^3 dy \int_{\frac{y}{2}}^y f(x, y) dx$$

$$\int_0^3 dy \int_{\frac{y}{2}}^y f(x, y) dx + \int_3^6 dy \int_{\frac{y}{2}}^{6-y} f(x, y) dx =$$



$$= \int_0^2 dx \int_x^{2x} f(x, y) dy + \int_2^3 dx \int_x^{6-x} f(x, y) dy$$

3496. $y^2 \leq 8x$, $y \leq 2x$, $y + 4x - 24 \leq 0$.

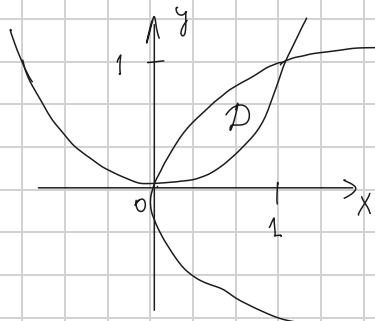


$$\int_0^2 dx \int_{-\sqrt{8x}}^{\sqrt{8x}} f(x, y) dy + \int_2^{4.5} dx \int_{-\sqrt{8x}}^{\sqrt{8x}} f(x, y) dy + \int_{4.5}^8 dx \int_{-\sqrt{8x}}^{4x+24} f(x, y) dy$$

3508. $\iint_D (x^2 + y) dx dy$, D — область, ограниченная параболлами $y = x^2$ и $y^2 = x$.

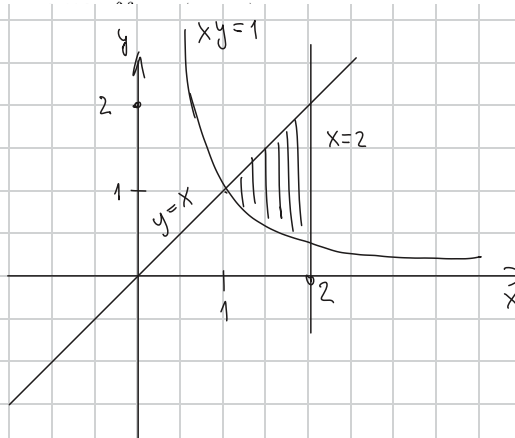
$$\iint_D (x^2 + y) dx dy = \int_0^1 dx \int_{x^2}^{\sqrt{x}} (x^2 + y) dy$$

$$\textcircled{1} \int_{x^2}^{\sqrt{x}} (x^2 + y) dy = \int_{x^2}^{\sqrt{x}} x^2 dy + \int_{x^2}^{\sqrt{x}} y dy = x^2 y \Big|_{x^2}^{\sqrt{x}} + \frac{y^2}{2} \Big|_{x^2}^{\sqrt{x}} = x^2 (\sqrt{x} - x^2) + \frac{1}{2} (x - x^4) = x^2 \sqrt{x} - x^4 + \frac{1}{2} x - \frac{1}{2} x^4 = -1.5x^4 + x^2 \sqrt{x} + 0.5x$$



$$\textcircled{2} \left(-1.5 \frac{x^5}{5} \right) \Big|_0^1 + \left(\frac{x^3 \sqrt{x}}{3.5} \right) \Big|_0^1 + \left(0.5 \frac{x^2}{2} \right) \Big|_0^1 = -0.3 + \frac{2}{7} + \frac{1}{4} = \frac{33}{140}$$

3509. $\iint_D \frac{x^2}{y^2} dx dy$, D — область, ограниченная прямыми $x = 2$, $y = x$ и гиперболой $xy = 1$.



$$\int_1^2 dx \int_{\frac{1}{x}}^x \frac{x^2}{y^2} dy$$

$$1) \int_{\frac{1}{x}}^x \frac{x^2}{y^2} dy = \left[-\frac{x^2}{y} \right]_{\frac{1}{x}}^x = -\frac{x^2}{x} - \left(-x^2 \cdot x \right) = x^3 - x$$

$$2) \int_1^2 (x^2 - x) dx = \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2 = \left(\frac{16}{3} - \frac{4}{2} \right) - \left(\frac{1}{3} - \frac{1}{2} \right) = \frac{9}{4}$$